

METRISCAN AI

PRODUCT REQUIREMENTS DOCUMENT (PRD)

AI-Assisted Legal Metrology Compliance Inspection Platform

SIH Problem Statement ID: 26034

Organization: Ministry of Consumer Affairs, Food & Public Distribution

Department: Department of Consumer Affairs (DoCA)

Category: Software

Document Version: 1.0 — MVP PRD

Purpose: Define exactly what MetriScan AI must do, who uses it, how the product behaves, what data it stores, and what constitutes a successful MVP for the Smart India Hackathon.

1. Document Overview

This PRD converts the SIH problem statement into an implementable software product specification. The system is an AI-assisted inspection platform, not an autonomous legal authority. AI is responsible for image understanding and extraction; a versioned deterministic compliance engine evaluates extracted information against applicable requirements; officers can review and override machine results with evidence.

Item	Definition
Product	MetriScan AI
Primary objective	Reduce the time and effort required to inspect packaged commodity labels.
Primary users	Enforcement officers, reviewers, administrators
Primary input	Photographs/images of packaged commodity labels and product information
Primary output	Structured declarations, compliance results, violations, evidence and reports
MVP principle	Human-in-the-loop, evidence-first, rule-versioned

2. Product Vision

Vision: Give enforcement officials a fast, explainable digital assistant that can inspect a packaged commodity from photographs, identify mandatory declarations, detect likely non-compliance, show the evidence behind each finding, and produce a standardized inspection report.

Product success means an officer can go from package photographs to a reviewable compliance report without manually reading every declaration.

3. Problem Statement → Product Response

Problem	MetriScan Response
Manual inspection is time-consuming	Automated image/OCR analysis and structured field extraction
Declarations may be missing	Mandatory-field detection and missing-field flags
Declarations may be incorrect/non-standard	Normalization and rule-based validation
Font/readability needs checking	OCR confidence, image quality and text-region analysis
Findings need evidence	Original image + bounding boxes + extracted value + rule reference
Inspection history is difficult to maintain	Central searchable repository and audit trail
Officials need monitoring	Dashboard with inspection and violation statistics
Reports are manual	Automated PDF and structured export

4. Personas & User Roles

Role	Needs	Permissions
Enforcement Officer	Fast inspection, evidence, reports, history	Create/manage own inspections; review; submit/finalize
Reviewer / Senior Officer	Review uncertain and high-risk findings	View assigned inspections; verify/edit findings; finalize
Administrator	Manage system and legal configuration	Users, roles, rules, rule versions, categories, audit logs

5. Primary User Journey

LOGIN → CREATE INSPECTION → ENTER PRODUCT DETAILS → UPLOAD/CAPTURE IMAGES → IMAGE QUALITY CHECK → OCR & EXTRACTION → FIELD REVIEW → COMPLIANCE CHECK → VIOLATION/EVIDENCE REVIEW → FINALIZE → GENERATE REPORT → SAVE TO HISTORY

Exception journey

If OCR confidence is low, the image is unreadable, a declaration is ambiguous, or a rule cannot be confidently applied, the result becomes **REVIEW** rather than an automatic FAIL. The officer/reviewer can inspect the evidence and correct the field.

6. Goals & Non-Goals

Goals

- Automate repetitive label-reading and declaration extraction.
- Detect missing or potentially non-compliant mandatory declarations.
- Provide explainable results with visual evidence.
- Maintain inspection history and auditability.
- Generate professional compliance/non-compliance reports.
- Provide an architecture that can later expand to additional regulatory domains.

Non-Goals for MVP

- Replacing the legal judgment of an authorized officer.
- Physically verifying actual net quantity using a weighing machine.
- Automatically issuing penalties or legal notices.
- Supporting every Indian language and every packaging type from the first release.
- Building a nationwide production-scale enforcement network during SIH.

7. Functional Requirements

ID	Requirement	Priority
FR-01	User shall be able to log in securely.	Must
FR-02	System shall enforce role-based permissions.	Must
FR-03	Officer shall create a new inspection.	Must
FR-04	Officer shall enter product/inspection metadata.	Must
FR-05	System shall accept one or multiple product images.	Must
FR-06	System shall validate image quality before analysis.	Must
FR-07	System shall preprocess images for OCR.	Must
FR-08	System shall extract text and bounding boxes using OCR.	Must
FR-09	System shall identify relevant declaration fields.	Must
FR-10	System shall attach confidence scores to extracted fields.	Must
FR-11	System shall normalize extracted values/units where possible.	Must
FR-12	System shall determine applicable compliance rules.	Must
FR-13	System shall evaluate mandatory declarations.	Must
FR-14	System shall identify missing/non-standard/suspicious declarations.	Must
FR-15	System shall support PASS/FAIL/REVIEW/NOT APPLICABLE.	Must
FR-16	System shall show evidence for each finding.	Must
FR-17	Officer shall be able to correct extracted fields.	Must
FR-18	System shall record corrections and reviewer identity.	Must
FR-19	System shall generate compliance reports.	Must
FR-20	System shall store inspection history.	Must
FR-21	System shall provide search/filter functionality.	Should
FR-22	System shall provide enforcement dashboard analytics.	Should
FR-23	Admin shall manage rule versions.	Must
FR-24	System shall maintain audit logs.	Must
FR-25	Reports shall be exportable to PDF and structured/editable formats.	Must

8. Functional Module Specifications

8.1 Authentication & RBAC

- Login with username/email and password.
- JWT/session-based authenticated API access.
- Roles: OFFICER, REVIEWER, ADMIN.
- Unauthorized users cannot access protected inspection/rule data.
- Audit authentication-related events.

8.2 Inspection Management

- Create inspection with inspection ID, date/time, location, officer and product metadata.
- Attach multiple images to one inspection.
- Maintain inspection lifecycle: DRAFT → ANALYZING → REVIEW → FINALIZED.
- Allow reopening only according to role permissions.

8.3 Image & Evidence Management

- Accept JPG/PNG and camera-captured images.
- Store original image without modification.
- Store processed image separately.
- Associate OCR bounding boxes with source image.
- Allow evidence preview from the inspection result.

8.4 OCR & Field Extraction

- Run OCR on relevant package regions.
- Return text, coordinates and confidence.
- Extract candidate fields such as manufacturer/packer/importer, commodity name, net quantity, MRP, dates, consumer-care details and other applicable declarations.
- Never silently discard low-confidence fields.

8.5 Compliance Engine

- Load applicable rule version based on inspection date/category/context.
- Check presence, format, consistency and other machine-checkable requirements.
- Return rule-level result with reason and evidence.
- Aggregate rule results into inspection-level status.
- Support REVIEW when evidence is insufficient.

8.6 Review & Correction

- Reviewer can edit extracted values.
- Every edit records previous value, new value, user and timestamp.
- Re-run affected compliance rules after a correction.
- Finalization locks the official inspection result unless reopened by authorized role.

8.7 Reporting

- Generate PDF report containing inspection details, product images, extracted declarations, findings, evidence, rule references and final status.
- Support structured export for downstream analysis.
- Give each report a unique identifier/version.

8.8 Repository & Search

- Search by inspection ID, product, date, officer, status and violation type.
- View historical inspection details.
- Keep previous report versions and audit history.

8.9 Dashboard

- Total inspections.
- Compliant / non-compliant / review counts.
- Violation category distribution.
- Inspection trend over time.

- Officer/activity summaries.
- High-priority review queue.

8.10 Rule Administration

- Create/update/deactivate rules.
- Maintain rule versions and effective dates.
- Record source/reference metadata.
- Prevent destructive modification of rules already used by finalized inspections.

9. Product Data Requirements

Inspection metadata

- Inspection ID
- Officer
- Date/time
- Location
- Product category
- Brand/product name
- Inspection status

Extracted declaration data

- Field name
- Raw OCR text
- Normalized value
- Unit
- Confidence
- Image ID
- Bounding box
- Verification status

Compliance result

- Rule ID
- Rule version
- Applicability
- Result
- Reason
- Evidence reference
- Reviewer status
- Timestamp

10. Status Model

Entity	Statuses
Inspection	DRAFT → ANALYZING → REVIEW → FINALIZED
Field	DETECTED → LOW_CONFIDENCE → VERIFIED / CORRECTED
Rule Result	PASS / FAIL / REVIEW / NOT_APPLICABLE
Report	GENERATING → READY / FAILED
User	ACTIVE / INACTIVE

11. Non-Functional Requirements

ID	Requirement	MVP Target
NFR-01	Usability	Officer should understand the main workflow without technical training.
NFR-02	Performance	Typical single-image analysis target: a few seconds to tens of seconds depending on hardware/model.
NFR-03	Reliability	Failures should be visible and recoverable; no silent data loss.
NFR-04	Security	Passwords hashed; protected APIs; role-based authorization.
NFR-05	Auditability	Changes to findings/rules/users recorded with timestamp and actor.
NFR-06	Explainability	Every machine-generated violation should show reason and evidence.
NFR-07	Scalability	Architecture should support multiple officers and growing inspection history.
NFR-08	Maintainability	Modular services, tests, documentation and environment configuration.
NFR-09	Data integrity	Finalized inspections and historical rule versions must remain reproducible.
NFR-10	Accessibility	Readable UI, clear status labels and responsive layout.

12. Compliance Decision Logic

1. Identify product category/context. 2. Select applicable rule version. 3. Determine which declarations are mandatory. 4. Map extracted fields to requirements. 5. Check presence. 6. Check machine-checkable format/value constraints. 7. Check readability/OCR confidence. 8. Produce rule-level results. 9. Aggregate into inspection status. 10. Route uncertain cases to human review.

Decision principle: missing evidence ≠ confirmed legal violation. If the system cannot reliably determine the state, use REVIEW.

13. AI / Computer Vision Requirements

Component	Input	Output	Acceptance Direction
Image quality	Original image	Quality score + warnings	Reject/flag unusable images
Preprocessing	Image	Enhanced/deskewed image	Improve OCR reliability
OCR	Processed image	Text + boxes + confidence	Accurate extraction on target labels
Field extraction	OCR result	Structured fields	High field-level precision on target dataset
Normalization	Structured candidate	Canonical value/unit	Equivalent representations unified
Confidence	Extraction + OCR signals	Confidence category	Low confidence routes to REVIEW

14. Rule Data Model Requirements

Each rule should be stored as structured configuration rather than buried inside application code.

```
Rule = { rule_id, rule_code, title, description, category, requirement_type, validation_logic, applicability_conditions, severity, effective_from, effective_to, source_reference, version, active }
```

Actual validation logic and legal references must be verified against the authoritative rules before being entered into production rule data.

15. Key User Stories & Acceptance Criteria

ID	User Story	Acceptance Criteria
US-01	As an officer, I want to log in securely.	Valid credentials grant access; invalid credentials are rejected; role is enforced.
US-02	As an officer, I want to create an inspection.	Inspection gets unique ID and stores officer/date/location/product data.
US-03	As an officer, I want to upload package images.	Images are stored, previewable and linked to the inspection.
US-04	As an officer, I want the system to analyze images.	OCR/extraction produces fields, confidence and evidence references.
US-05	As an officer, I want to see missing declarations.	Applicable missing fields are explicitly flagged.
US-06	As an officer, I want to know why something failed.	Each failure shows rule, reason and evidence.
US-07	As a reviewer, I want to correct OCR errors.	Correction is saved, audited and can trigger rule re-evaluation.
US-08	As a reviewer, I want uncertain cases separated.	REVIEW cases are clearly identified and filterable.
US-09	As an officer, I want a report.	Report contains inspection details, findings, evidence and final status.
US-10	As an administrator, I want versioned rules.	Rule versions have effective dates and historical inspections remain reproducible.
US-11	As an official, I want historical searches.	Search/filter returns relevant past inspections and reports.
US-12	As an administrator, I want dashboard statistics.	Counts/trends reflect stored inspections and violation results.

16. Frontend Screen Requirements

Screen	Purpose	Major Components
Login	Authenticate user	Email/username, password, role-aware login
Dashboard	Monitor activity	KPIs, charts, recent inspections, review queue
New Inspection	Start workflow	Product metadata, category, location, image upload
Image Review	Check input	Image preview, quality warnings, retake/upload
Analysis Progress	Show processing	OCR/extraction progress and errors
Compliance Results	Main decision screen	Overall status, field table, rule results, confidence
Evidence View	Explain findings	Image with bounding boxes, OCR text, rule explanation
Review/Edit	Human correction	Editable fields, previous/current value, save/re-run
Inspection History	Search repository	Filters, status, dates, product, officer
Report Preview	Finalize output	Report summary, evidence, export buttons
Rule Admin	Manage legal configuration	Rules, versions, effective dates, activation
User Admin	Manage access	Users, roles, active/inactive status

17. Core API Requirements

Example API contract

```
POST /api/inspections/{inspection_id}/analyze Response: { "status": "REVIEW", "fields": [
{"name":"mrp","value":"■120","confidence":0.97}, {"name":"net_quantity","value":"500
g","confidence":0.98} ], "violations": [], "review_required": true }
```

The actual API specification will be expanded during the System Architecture stage into request schemas, response schemas, authentication rules and error codes.

18. Database Requirements

Core entities: USER, ROLE, INSPECTION, PRODUCT, IMAGE, OCR_RESULT, EXTRACTED_FIELD, RULE, RULE_VERSION, RULE_RESULT, VIOLATION, REPORT, AUDIT_LOG

Key relationship: one inspection can contain multiple images and extracted fields; each rule result belongs to an inspection and a specific rule version; evidence links results back to image regions.

19. Security & Privacy Requirements

- Use HTTPS in deployed environments.
- Never store plain-text passwords.
- Apply role-based authorization at API level, not only in the UI.
- Validate uploaded file types and size limits.
- Store uploaded files outside executable application paths.
- Restrict access to inspection data according to user role.
- Maintain audit logs for edits, finalization and rule changes.
- Do not expose secrets in frontend code or Git repositories.

20. Compliance Report Requirements

Report should contain: report ID, inspection ID, date/time, officer, location, product details, image/evidence summary, extracted declarations, rule-level findings, violation descriptions, confidence/review notes, final status and reviewer/finalizer details.

21. Dashboard KPI Requirements

- Total inspections completed.
- Compliance rate.
- Non-compliance count.
- Review-pending count.
- Top violation categories.
- Inspections by date/time period.
- Inspections by product category.
- Officer activity.
- Average analysis/review time.

22. Edge Cases

Scenario	Expected Behavior
Blurred image	Warn user and request retake/re-upload.
Rotated package	Attempt orientation correction before OCR.
Multiple declarations on different sides	Allow multiple images and combine evidence.
OCR cannot read field	Set low confidence/REVIEW rather than invent value.
Duplicate upload	Warn or allow duplicate with clear linkage.
Rule not applicable	Return NOT_APPLICABLE with explanation.
Rule version unavailable	Block finalization and show configuration issue.
Network/API failure	Preserve draft and allow retry.
Reviewer changes extracted value	Audit old/new values and re-run affected checks.
Finalized inspection	Read-only unless authorized reopening.

23. MVP Definition of Done

- User can authenticate and role permissions work.
- Officer can create an inspection and upload package images.
- System performs image preprocessing and OCR.
- System extracts a defined MVP set of mandatory declaration fields.
- Each extracted field has confidence and evidence.
- Rules are stored separately from application code and are versioned.
- At least a meaningful initial rule set is implemented and tested against verified examples.
- System produces PASS/FAIL/REVIEW/NOT APPLICABLE rule results.
- Officer/reviewer can inspect evidence and correct fields.
- PDF report is generated successfully.
- Inspection history is searchable.
- Dashboard displays stored inspection statistics.
- Core APIs have automated tests.
- Demo can run end-to-end on a prepared dataset.

24. Success Metrics

Metric	MVP Measurement
Field extraction accuracy	Measure against manually verified ground truth.
OCR quality	Measure character/field accuracy on target package images.
Violation detection precision	Measure confirmed violations against labeled test cases.
Review rate	Track percentage requiring human review.
End-to-end processing time	Measure upload-to-result duration.
Report generation success	Percentage of finalized inspections producing valid reports.
Usability	Officer/judge can complete demo workflow without developer intervention.

25. Risks & Mitigations

Risk	Impact	Mitigation
OCR errors	Wrong extraction	Preprocessing, confidence scores, human review, multiple images
Legal rule ambiguity	Incorrect compliance result	Versioned rule model + authoritative source verification + REVIEW state
Insufficient dataset	Poor AI performance	Build controlled golden dataset and synthetic edge cases
Font-size/readability complexity	Hard to automate reliably	Treat as assisted evidence/check rather than unsupported exact legal judgment
Scope creep	MVP not finished	Limit initial categories/fields/rules
Overengineering	Development delays	Modular monolith first
AI hallucination	Unsafe findings	Never allow generated text to become unsupported legal fact

26. Release Plan

Release	Scope
Prototype 0.1	Image → OCR → fields → simple rule checks in Python
MVP 0.5	Backend + database + frontend + inspection workflow

Release	Scope
MVP 0.9	Evidence, review, reporting, dashboard and audit logs
SIH Demo 1.0	Stable end-to-end demo, polished UI, prepared dataset, tests and deployment

27. Final Product Requirement

MetriScan AI shall provide a secure, explainable and auditable workflow in which an authorized user can submit packaged commodity images, receive structured declaration extraction with confidence/evidence, evaluate those declarations against the applicable versioned compliance rules, review or correct uncertain findings, preserve the inspection history, and generate a compliance/non-compliance report.

Next document: System Architecture Specification — component architecture, deployment architecture, data flow, sequence diagrams, ER diagram, database schema, API architecture, AI pipeline architecture, rule-engine architecture and frontend/backend folder structure.